SIMATS ENGINEERING

CSA6303- Threat Intelligence and Network Security

A1-Threat Intelligence Report

Assessment Questions

Subject	Threat Intelligence	**Total Marks**	50
Questions	5 (all compulsory)	**Marks per Question**	10
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Instructions: Answer all questions. Each question carries 5 marks. Support answers with examples and diagrams where appropriate.

CO3: Perform threat analysis using intelligence models and utilize industry tools for monitoring and detection.

Code of Conduct:

I T. Bhanu Pavan Varma(192425380) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of Student

Questions :

1. **Threat Hunting Techniques:** An organization suspects that an attacker has gained unauthorized access to its internal network. The security team observes unusual login times, unexpected processes, and connections to unfamiliar external servers. Describe the proactive threat-hunting techniques you would use to investigate the activity and identify potential threats. [10]

2. **SIEM Tools and Security Monitoring :** A company implements a Security Information and Event Management (SIEM) system to monitor its network. The system generates multiple alerts about suspicious login attempts and unusual outbound connections. Explain how the SIEM system can collect and analyze security events, correlate alerts, and help security analysts prioritize potential threats.
   [10]

3. **Security Monitoring and Anomaly Detection:** A company's security monitoring dashboard identifies a sudden increase in outbound network traffic during non-working hours. The activity originates from a database server that normally transfers only small amounts of data. Explain how you would investigate the anomaly, identify possible data exfiltration, and recommend suitable monitoring and prevention measures. [10]

4. **Tools Used in the IT Industry for Threat Hunting:** An IT organization suspects that malware has infected several employee workstations. The security team has access to SIEM software, endpoint detection tools, network traffic analyzers, and vulnerability scanners. Select suitable tools for investigating the incident and explain how each tool can help identify, analyze, and contain the threat.
   [10]

5. **Integrated Threat Hunting and Incident Reporting:** A company detects suspicious user logins, connections to a malicious domain, and an unknown process running on an employee workstation. The security team uses Splunk or the ELK Stack to analyze logs and a threat intelligence platform to verify suspicious indicators. Describe the complete threat-hunting process, including data collection, log correlation, threat investigation,

malware analysis if required, risk assessment, and preparation of a threat-hunting report with recommended remediation actions. [10]

Q1. Threat Hunting Techniques (10 Marks)

Definition: Threat hunting is a proactive security process in which analysts search for hidden attackers, suspicious activities, and security threats that automated tools may have missed.

Techniques used to investigate the activity

1. Log Analysis: Examine authentication, firewall, DNS, and system logs to identify unusual login times, repeated failures, and suspicious connections.

2. Endpoint Investigation: Use Endpoint Detection and Response (EDR) tools to inspect running processes, startup programs, and unauthorized changes.

3. Behavioral Analysis: Compare user and system activities with normal behavior to identify unusual login locations, times, or access patterns.

4. Network Traffic Analysis: Use Wireshark or Zeek to investigate unfamiliar external connections, unusual protocols, and suspicious data transfers.

5. Threat Intelligence Analysis: Check suspicious IP addresses, domains, and file hashes against trusted threat intelligence sources.

6. Process and Memory Analysis: Examine suspicious processes and, when justified, collect memory evidence to identify malware or unauthorized execution.

7. Lateral Movement Detection: Search for unusual connections between internal systems and suspicious use of administrative accounts.

8. Hypothesis-Based Hunting: Develop hypotheses such as “an attacker is using a compromised account” and test them against available evidence.

9. Vulnerability Assessment: Scan authorized systems for weaknesses that could have enabled unauthorized access.

10. Containment and Reporting: Isolate confirmed compromised endpoints, secure affected accounts, preserve evidence, and document findings.

Conclusion: Threat hunting combines log analysis, endpoint monitoring, network investigation, and threat intelligence to detect and respond to hidden threats before they cause greater damage.

Q2. SIEM Tools and Security Monitoring (10 Marks)

Definition: Security Information and Event Management (SIEM) is a security solution that collects, centralizes, correlates, and analyzes logs and events from multiple sources to detect threats.

How SIEM works

1. Data Collection

Firewall, servers, endpoints, applications, authentication logs

2. Log Normalization and Processing

Convert events into a consistent format and enrich them with context

3. Correlation and Analysis

Connect suspicious logins, endpoint activity, and outbound connections

4. Alerting and Prioritization

Generate alerts based on severity, threat context, and asset importance

5. Investigation and Response

Analysts investigate incidents, contain threats, and document results

How SIEM helps security analysts

1. Centralized monitoring: Collects logs from multiple systems in one location.
2. Event correlation: Links related events, such as repeated login failures followed by a successful login.
3. Threat detection: Identifies suspicious patterns using rules, behavioral analytics, and threat intelligence.
4. Alert prioritization: Assigns severity based on suspicious behavior, potential impact, and affected assets.
5. Real-time alerting: Notifies analysts when suspicious events meet detection rules.
6. Incident investigation: Provides timelines and related events for tracing attacker activity.
7. Compliance and reporting: Supports audit trails and security reports.
8. Automated response: Can integrate with security orchestration tools to block malicious IPs or disable compromised accounts, subject to configured safeguards.

Examples of SIEM tools: Splunk Enterprise Security, Microsoft Sentinel, and Elastic Security.

Conclusion: SIEM improves visibility across an organization's network and helps analysts detect, investigate, and prioritize potential threats efficiently.

Q3. Security Monitoring and Anomaly Detection (10 Marks)

The sudden increase in outbound traffic from a database server during non-working hours may indicate data exfiltration, malware activity, unauthorized backups, or a legitimate scheduled process.

Investigation steps

1. Validate the alert: Compare the traffic volume with historical baselines and check whether scheduled jobs explain the activity.

2. Analyze network logs: Review firewall, NetFlow, DNS, and proxy logs to identify destination IP addresses, domains, ports, protocols, and transfer volumes.

3. Identify the destination: Check whether the external server is unfamiliar, suspicious, or associated with known malicious infrastructure.

4. Examine database activity: Review database audit logs for unusual queries, bulk exports, unauthorized access, or unexpected account usage.

5. Inspect the server: Investigate running processes, active connections, scheduled tasks, and recently modified files.

6. Correlate events: Match network activity with user logins, database queries, and endpoint alerts to establish a timeline.

7. Check for exfiltration: Look for unusually large transfers, compressed archives, unexpected encrypted connections, or repeated transfers to external destinations.

8. Assess the risk: Determine which data may have been exposed, the sensitivity of that data, and whether the activity is ongoing.

9. Contain the incident: If malicious activity is confirmed or strongly suspected, restrict suspicious outbound connections and isolate the server when operationally safe. Preserve logs and evidence.

10. Report and recover: Notify the incident response team, remediate the cause, restore systems if necessary, and monitor for recurrence.

Monitoring and prevention measures

- Data Loss Prevention (DLP): Detect or restrict unauthorized transfer of sensitive data.
- Egress filtering: Allow outbound connections only to approved destinations and services.
- Network monitoring: Use NetFlow, Zeek, or IDS tools to detect unusual traffic.
- Database auditing: Monitor bulk queries, privileged access, and unexpected exports.
- SIEM alerts: Establish thresholds and behavioral rules for abnormal traffic volumes.
- Encryption and access control: Protect sensitive data and limit unauthorized access.
- Incident response procedures: Maintain tested procedures for containment and recovery.

Conclusion: Correlating network traffic, database logs, and endpoint activity helps determine whether the anomaly represents data exfiltration and supports timely containment.

Q4. Tools Used in the IT Industry for Threat Hunting

When malware is suspected on employee workstations, the security team should combine endpoint, SIEM, network, threat intelligence, and vulnerability assessment tools.

Tool	Category	Purpose in threat hunting
Splunk	SIEM	Correlates logs and detects suspicious login patterns and events.
Elastic Security (ELK Stack)	SIEM and analytics	Searches and correlates logs to build incident timelines.
Microsoft Defender for Endpoint	EDR	Investigates processes, detects malware, and supports endpoint isolation.
CrowdStrike Falcon	EDR	Provides endpoint telemetry, threat detection, and response capabilities.
Wireshark	Packet analyzer	Inspects network packets and suspicious communications.
Zeek	Network security monitoring	Produces network logs about connections, DNS, and protocol activity.
Nessus	Vulnerability scanner	Identifies weaknesses that may have enabled the infection.
OpenVAS	Vulnerability scanner	Discovers known vulnerabilities and misconfigurations.
VirusTotal	Threat intelligence analysis	Checks file hashes, URLs, and domains against multiple security engines and datasets.
YARA	Malware identification	Matches files or memory artifacts against rules describing suspicious characteristics.

How the tools work together

1. Detect: SIEM tools identify suspicious logins and security events.
2. Investigate: EDR tools examine processes, files, and endpoint behavior.
3. Analyze: Wireshark and Zeek investigate network communications.
4. Verify: Threat intelligence tools check suspicious indicators.
5. Assess: Nessus or OpenVAS identifies exploitable weaknesses.
6. Contain: EDR or approved security controls isolate infected endpoints and block confirmed malicious indicators.
7. Recover: Remove malware, patch vulnerabilities, reset compromised credentials, and monitor systems.

Conclusion: Integrating these tools gives security teams better visibility, faster threat detection, and more effective incident containment.

Q5. Integrated Threat Hunting and Incident Reporting

Scenario: A company detects suspicious user logins, connections to a malicious domain, and an unknown process on an employee workstation. The team uses Splunk or the ELK Stack and a threat intelligence platform to investigate.

Complete threat-hunting process

1. Define the hunting objective

- Establish the suspected incident and the affected user or workstation.
- Form a hypothesis that a compromised account or malware may be responsible.

2. Collect security data

Gather relevant information from:

- Authentication and identity logs.
- Endpoint and process execution logs.
- DNS, firewall, proxy, and network connection logs.
- EDR alerts and threat intelligence feeds.

3. Correlate logs using SIEM

Use Splunk or Elastic Security to correlate events by timestamp, user identity, host, IP address, and domain.

For example, a suspicious login followed by an unknown process and a connection to a malicious domain may indicate a coordinated attack.

4. Verify suspicious indicators

- Check domains, IP addresses, and file hashes against trusted threat intelligence sources.
- Review reputation, related indicators, and known attacker techniques.
- Treat threat intelligence matches as evidence to investigate, not automatic proof of compromise.

5. Investigate the endpoint

- Examine process trees, command-line activity, persistence mechanisms, and file modifications.
- Determine how the process started and whether it communicated with suspicious external servers.
- Preserve forensic evidence before making changes when practical.

6. Perform malware analysis if required

- Calculate the suspicious file's hash and check available reputation data.
- Use an isolated analysis environment to examine its behavior when necessary.
- Identify persistence, network communication, and potential impact without exposing production systems.

7. Assess risk and scope

- Identify affected devices, accounts, and sensitive data.
- Determine whether the attacker moved laterally to other systems.
- Prioritize response according to confidence, exploit activity, asset importance, and business impact.

8. Contain and eradicate

- Isolate confirmed compromised endpoints.
- Disable or secure compromised accounts and revoke active sessions.
- Block verified malicious domains and indicators.
- Remove malicious artifacts and remediate the initial entry point.

9. Recover and monitor

- Restore systems to a trusted state.
- Apply patches and strengthen access controls.
- Monitor for recurring indicators and suspicious activity.

10. Prepare the threat-hunting report

The report should contain:

Section	Contents
Executive summary	Incident overview and overall risk
Scope	Systems, users, and time period investigated
Evidence	Relevant logs, timestamps, hashes, IPs, and domains
Findings	Confirmed facts and suspected attacker behavior
Attack timeline	Chronological sequence of observed events
Risk assessment	Likelihood, impact, and affected assets
Response actions	Containment, eradication, and recovery steps
Recommendations	Patching, MFA, monitoring, and access-control improvements

Conclusion: Integrated threat hunting combines log correlation, threat intelligence, endpoint investigation, and risk assessment to identify threats, determine their impact, contain incidents, and improve the organization's future security posture.